MAGNETISM

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But any degree of temper that is given to feed makes a very important change in this respect. In the first place, it acquires magnentifin more flowly by induction than an equal and similar piece of common iron, and its maly acquired lets. These differences are only trans- mined by the deviations which it causes in the manners needle from the magnetic meridian, and by its attractive tions.

41 tempered sheet from a the magma film.

When the inducing magnet is removed, fome mag- netti remains in the fixed bar, which retains the polory which, it will be the neighborhood of the matter.

Steel tempered to the degree fit for watch Springs acquires along magnetic which is exhibits imme disels on the removal of the magnet. But it diffipates very fat ; and, in a very few minutes, it is reduced to lois than one half of its strength while in control with the magnet, and not two thirds of what it was distety on removal from it. It continues to diffidate for former days, though the her be kept with care'; but the diffusion diminishes at, and in the near at, half over third of its greatest power forany length of time, unleef cardially keep or indjudicially treated.

Steel tempered into flowing cutting tools, fish a schlieb, punches, and drills for metal, acquires magnifin fill more flowly by induction, and acquires five of it while in contact with the magnet, but it certain it more firms by, and finally retains a greater proportion of who it had acquired.

Steel made as hard as possible, is much longer in ac- quiring all the magnifin which limple juxtaposition can give to it. It acquires less than the formerly but it retains is with great firmness, and finally retains a much greater proportion.

Such one of iron as are fulcypible of magnetifi, are nearly like hard feel in their effects you that in the time necessary for their _properlift_ impregnation, and in the durability of the acquired magnifine. They effic exceedingly in respect to the degree of power which they can attain by mere juxtaption, and the virieties form to depend on heterogeneous mixture. We emit offriever, thus far once of our ones of influence of of magnetism of the metal in the fate of my way, and com- bred with dolphins, are not magnentable while remain- ing in that there. Mott core require reading, and a fort of cementation, in contact with inflammable fuhlances. This matter is not underdefended, but it would form that complex metallization is far from being the north to votrale condition, and that a certain degree of oxyda- tion, and perhaps some other positive yet unknown, make the her lautiness. Our all this is extremely oblicure. The late Dr Cowin Knight made a compolio- tion which required every phono and permanent mag- seting, but the recent died with him. Dr Gilbert peaks of limilat compositions, in which ferruginous days were ingredients ; but we know nothing of the fate of the metal in them, nor their mode of acquiring mag- retin.

It is of peculiar importance to work that the ac- quition of magnetization is gradual and progrestive, and that the gradation is the more perceptible in proportion as the field is of a hastheric temper. When a magnet is hrough to one end of a bar of common iron, to remote extremity, unleef exceedingly long, requires its utmost magnifin immediately. But when the north pole of

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a magnet is applied: to one end of a bar of hard flee, the part in contact immediately becomes a fourth pole, and the far end is not yet affected. We offer a north pole formed a fance difference from the concept, and bec- sent that a faint tooth pole. "Their gradually advance along the bar. The remote extremity become frit a faint fourth pole, and it is not differ very long while (If ceer) that it becomes a simple, vigorous, north pole. More frequently it remains a difficult and recycle north pole: may, if the her for very long, it often hop- pus that have a sacrifice of north and forth poles, which never make their way to the far end of the bar. This phenomenon was first observed (we think) by Dr Brook Taylor, who _they_ now account of his obtivations to the _Philophysical Transformation_, no 344.

From the account we have given of their phenomenons to a st- of indeed magnifine, it appears that the temporary because magnifin is always lo disfpided that the rim of their beces mutual attractions of the difficiliate poles exceeds the gentich im of the repollon between the limlar, poles, and that therefore the two magnets tend to each other. This is evidently equivalent to flying, that a piece of magnungtic iron is always attracted by a magnet. No exception has ever been observed to this fact, for Phys's dury of a Thamades, or loadlines, which repeals iron, is allowed by all to have been a false.

We think outfellow authorified to lay that this attraction of the headlines for iron, or this tendency of iron to the headline, is a secondary phenomenon, and is the _conference_ of the proper disposition of the induced mag- netimin. The proofs already given of the compound matters of this phenomena, namely, that is arise from the excess of two attributions above two repetitions, need (we imagine) no addition. But the following condir- actions place the matter beyond doubt.

I. The magnifinal of the two poles is evidently of an opposite nature; the one repelling what the other at- trade). If the aircraft iron, therefore, the other should result it. But each path, by changing a magretifion opposite to its own, on the nearest end of the iron, and the fans with its own on the remote end, and its action diminishing with an increase of diflants, there mult always be an excess of attraction, and the iron mult be attached.

A Each of the magnets A and B, in either of the positions reprefened in fig. 12. would alone attract the piece of common iron. But when placed together, the fourth pole of 8 tends to render the upper end of C a north pole; while the north pole of B tends to take it a South pole. If there actions be nearly equal, the weight of C cannot be supposedly be the magnifin induced by any difference of action that may remain. While C is hanging D I alone, let. A be gradually enough tree, it gradually thorley the action of the north pole of B, to that C gradually lofos its magnifin and polarity, and its weight prends.

ğ. He all three cases where the induction of magne- tifism is flow, the attraction is weak in proportion. This is particularly remarked by Dr Gilbers. If we take piece of common iron, and of fleet of different temps, but all of the fame fine and form, we shall find that the iron is much more thought attracted than any of the rel, and that the attraction for each of them a weaker proportion as they are harder. This liberty is to accurately observed, that when the piece is thoroughly

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43 Tempered steel acquires durable magnetism.

But any degree of temper that is given to ſteel makes a very important change in this reſpect. In the firſt place, it acquires magnetiſm more ſlowly by induction than an equal and ſimilar piece of common iron, and finally acquires leſs. Theſe differences are eaſily examined by the deviations which it cauſes in the mariner's needle from the magnetic meridian, and by its attraction.

When the inducing magnet is removed, ſome magnetiſm remains in the ſteel bar, which retains the polarity which it had in the neighbourhood of the magnet.

Steel tempered to the degree fit for watch ſprings acquires a ſtrong magnetiſm, which it exhibits immediately on the removal of the magnet. But it diſſipates very faſt; and, in a very few minutes, it is reduced to leſs than one-half of its intenſity while in contact with the magnet, and not two-thirds of what it was immediately on removal from it. It continues to diſſipate for ſome days, though the bar be kept with care; but the diſſipation diminiſhes faſt, and it retains at leaſt one-third of its greateſt power for any length of time, unleſs careleſsly kept or injudiciouſly treated.

Steel tempered for ſtrong cutting tools, ſuch as chiſels, punches, and drills for metal, acquires magnetiſm ſtill more ſlowly by induction, and acquires leſs of it while in contact with the magnet; but it retains it more firmly, and finally retains a greater proportion of what it had acquired.

Steel made as hard as poſſible, is much longer in acquiring all the magnetiſm which ſimple juxtapoſition can give to it. It acquires leſs than the former; but it retains it with great firmneſs, and finally retains a much greater proportion.

Such ores of iron as are ſuſceptible of magnetiſm, are nearly like hard ſteel in theſe reſpects; that is, in the time neceſſary for their *greateſt* impregnation, and in the durability of the acquired magnetiſm. They differ exceedingly in reſpect to the degree of power which they can attain by mere juxtapoſition, and the varieties ſeem to depend on heterogeneous mixture. We muſt obſerve, that few ores of iron are ſuſceptible of magnetiſm in their natural ſtate. The ordinary ores, conſiſting of the metal in the ſtate of an oxyd, and combined with ſulphur, are not magnetizable while remaining in that ſtate. Moſt ores require roaſting, and a ſort of cementation, in contact with inflammable ſubſtances. This matter is not well underſtood; but it would ſeem that complete metallization is far from being the moſt favourable condition, and that a certain degree of oxydation, and perhaps ſome other compoſition, yet unknown, make the beſt loadſtones. But all this is extremely obſcure. The late Dr Gowin Knight made a compoſition which acquired a very ſtrong and permanent magnetiſm, but the ſecret died with him. Dr Gilbert ſpeaks of ſimilar compoſitions, in which ferruginous clays were ingredients; but we know nothing of the ſtate of the metal in them, nor their mode of acquiring magnetiſm.

44 Induction magnetism is gradual and progreſſive.

It is of peculiar importance to remark that the acquiſition of magnetiſm is gradual and progreſſive, and that the gradation is the more perceptible in proportion as the ſteel is of a harder temper. When a magnet is brought to one end of a bar of common iron, its remote extremity, unleſs exceedingly long, acquires its utmoſt magnetiſm immediately. But when the north pole of a magnet is applied to one end of a bar of hard ſteel, the part in contact immediately becomes a ſouth pole, and the far end is not yet affected. We obſerve a north pole formed at ſome diſtance from the contact, and beyond this a faint ſouth pole. Theſe gradually advance along the bar. The remote extremity becomes firſt a faint ſouth pole, and it is not till after a very long while (if ever) that it becomes a ſimple, vigorous, north pole. More frequently it remains a diffuſed and feeble north pole: nay, if the bar be very long, it often happens that we have a ſucceſſion of north and ſouth poles, which never make their way to the far end of the bar. This phenomenon was firſt obſerved (we think) by Dr Brook Taylor, who gives an account of his obſervations in the *Philoſophical Tranſactions*, n° 344.

45 Iron is attracted only because it becomes magnetical.

From the account we have given of theſe phenomena of induced magnetiſm, it appears that the temporary magnetiſm is always ſo diſpoſed that the ſum of the mutual attractions of the diſſimilar poles exceeds the ſum of the repulſions between the ſimilar poles, and that therefore the two magnets tend to each other. This is evidently equivalent to ſaying, that a piece of unmagnetic iron is always attracted by a magnet. No exception has ever been obſerved to this fact; for Pliny's ſtory of a Theamedes, or loadſtone, which repels iron, is allowed by all to have been a fable.

We think ourſelves authoriſed to ſay that this attraction of the loadſtone for iron, or this tendency of iron to the loadſtone, is a ſecondary phenomenon, and is the *conſequence* of the proper diſpoſition of the induced magnetiſm. The proofs already given of the compound nature of this phenomenon, namely, that it ariſes from the exceſs of two attractions above two repulſions, need (we imagine) no addition. But the following conſiderations place the matter beyond doubt:

1. The magnetiſm of the two poles is evidently of an oppoſite nature; the one repelling what the other attracts. If the one attracts iron, therefore, the other ſhould repel it. But each pole, by inducing a magnetiſm oppoſite to its own, on the neareſt end of the iron, and the ſame with its own on the remote end, and its action diminiſhing with an increaſe of diſtance, there muſt always be an exceſs of attraction, and the iron muſt be attracted.

2. Each of the magnets A and B, in either of the poſitions repreſented in fig. 12. would alone attract the piece of common iron C. But when placed together, the ſouth pole of A tends to render the upper end of C a north pole; while the north pole of B tends to make it a ſouth pole. If their actions be nearly equal, the weight of C cannot be ſupported by the magnetiſm induced by any difference of action that may remain. While C is hanging by B alone, let A be gradually brought near; it gradually deſtroys the action of the north pole of B, ſo that C gradually loſes its magnetiſm and polarity, and its weight prevails.

3. In all thoſe caſes where the induction of magnetiſm is ſlow, the attraction is weak in proportion. This is particularly remarked by Dr Gilbert. If we take pieces of common iron, and of ſteel of different tempers, but all of the ſame ſize and form, we ſhall find that the iron is much more ſtrongly attracted than any of the reſt, and that the attraction for each of them is weaker in proportion as they are harder. This diverſity is ſo accurately obſerved, that when the piece is thoroughly

Figure 1: *
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